CARA contacts-off trajectory validation: team guide

Status: prepared for local execution; no new test has been run. Numerical findings and videos will be generated by the user's run. This document describes the protocol and how to interpret the resulting evidence.

1. Objective

Test whether the supplied analytical FK, IK and Jacobian agree with a fixed-base, single-leg CAD-derived MuJoCo model along a specified path. The two phase results are saved before comparison. No analytical equations are used to drive the reference baseline.

The scope is ideal geometric agreement for this trajectory and assembly branch. Physical accuracy, collisions, motor torque requirements, balance and real-time control are outside this experiment.

2. Trajectory and sampling

The foot travels once around a base circle, then follows four rising turns with a linearly shrinking radius as a function of a smoothed progress variable. Center: (-80, 65, -240) mm. Radius: 35 mm. Rise: 60 mm. Apex: (-80, 65, -180) mm. Nominal base duration: 4 s; rise duration: 12 s.

The progress function is s(u) = 10u^3 - 15u^4 + 6u^5. During the rising portion, x = cx + R(1-s)cos(theta), y = cy + R(1-s)sin(theta), z = cz + Hs, where theta = 2pi + 2pi*turns*s. The base circle uses theta = 2pi*s(t/4).

Default sampling increases from 401 to 1,601 points, giving 0.01 s nominal spacing and 400 intervals around the base circle. All points are stored, regardless of video frame rate. This increases resolution on the same path; it does not independently test additional workspace regions or branches. Samples are spatially correlated, not 1,601 independent physical experiments.

3. Baseline phase

Enable the equality connecting FL_P to the movable mocap target; disable independent-angle holds. Move the target through a 60-point approach from home, then through every requested trajectory point. MuJoCo steps its constraint dynamics until the configured settling criteria or maximum step count.

At each point store desired XYZ, actual foot XYZ, three actual independent angles, full qpos/qvel, simulated time, speed, loop closure gap, branch indicator, joint-limit violation and status. The baseline NPZ includes the corresponding desired trajectory and nominal timestamps.

Independent-angle and foot-target constraints are never simultaneously used to drive a baseline sample. Gravity and contacts are off, and damping is numerical. This is constraint relaxation, not a motor-driven gait controller.

4. Analytical phase

Use the same desired XYZ points, not the achieved baseline XYZ. Your analytical IK supplies the three angle requests. Disable the foot-target equality and enable only independent-angle holds, retaining the loop equality.

MuJoCo settles the linkage and measures the resulting foot position. The red target marker remains visible but cannot pull the foot in this mode. Record the same state and diagnostic fields, plus requested angles and analytical FK(IK(target)) self-consistency error.

Only a successfully held pose updates the previous-solution seed. An analytical exception is recorded as IK_NO_SOLUTION; its saved pose is unavailable. An existing mathematical solution with poor settling is recorded as INVALID_POSE. Neither is silently discarded. The equations and retained home-angle calibration are unchanged from the supplied project.

5. Offline comparison

The comparison verifies complete output counts, matching source/config/environment signatures and hashes of saved sample/state files. It loads both saved runs. It does not run physics or solve IK again. MuJoCo forward calculations and site Jacobians are evaluated at saved baseline configurations.

Compare: baseline tracking error; analytical FK versus baseline at identical measured motor angles; achieved analytical IK foot versus desired target; achieved positions between the two phases; analytical versus CAD effective Jacobian; and predicted velocities for the same rates (5, 10, -7) deg/s. Also record wrapped measured-angle differences, which are descriptive and not proof of a unique IK solution.

The CAD effective Jacobian eliminates passive rates. If Aa*qa_dot + Ap*qp_dot = 0, solve Ap*B = -Aa and compute J_eff = Ja + Jp*B. The reported Jacobian error is the largest absolute entry difference. The velocity comparison is instantaneous prediction from the two Jacobians, not independent finite-difference video tracking.

6. Error and acceptance definitions

Position error = Euclidean norm of the relevant XYZ difference, in mm. Jacobian error = maximum entry difference, in mm/rad. Velocity error = Euclidean norm of velocity difference, in mm/s. RMS = sqrt(mean(error squared)). Summaries also report count, median and 95th percentile.

Engine requirements: position tracking < 0.1 mm in target mode; input-angle tracking < 0.01 degrees in angle mode; loop closure < 0.05 mm; maximum speed < 0.001 rad/s; home assembly sign retained; any defined limits satisfied within angle tolerance. The loop velocity residual must be below 1e-7 m/rad for Jacobian reference eligibility.

Comparisons have separate thresholds: FK and IK maximum error <= 0.1 mm, Jacobian maximum entry error <= 0.01 mm/rad, and predicted velocity error <= 0.01 mm/s. These are configurable numerical test thresholds, not experimental hardware specifications. A comparison passes only with full valid coverage, no missing metric entries, and its maximum below threshold.

Small finite errors at invalid poses do not count as successful validation. The stage 'complete' status means the workflow finished, not that the derivations passed. Read summary.json acceptance and validity counts. No physical limits were provided, so a zero limit violation does not establish real motor feasibility.

7. Recording and evidence

Each phase writes actual transient simulation capture to physics_settling.mp4, with simulation-time captions, plus transient frame states in physics_frames.npz. At the defaults, one frame is captured every 33 physics steps and encoded at 30 fps: about 0.99 simulated seconds per video second. Baseline approach is included and labeled sample -1.

A separate trajectory_playback.mp4 displays nearest saved poses on the nominal 16-second timeline. It is labeled playback and does not integrate new dynamics. Videos do not necessarily display every numerical sample. Closed-loop consistency comes from the saved states, not image interpolation.

Both recordings use an offscreen MuJoCo renderer; a working OpenGL driver and FFmpeg encoder are required. Failure to render or encode marks the stage incomplete. Numerical checkpoints are saved periodically. Missing IK states are explicitly labeled during playback. The recordings are not desktop screenshots or a physical tracking-performance claim.

8. Reproduction and inspection

Use Python 3.12 in a new virtual environment and install requirements.txt. Run check_setup.py locally; it loads the model and checks rendering without stepping physics. Then run workflow.py --stage all --out results/run_1601

--preview. Omit --preview to hide the viewer while keeping offscreen recordings. Windows commands are in README.md.

Alternatively run baseline, analytical and compare as separate stages using the same output path. The compare stage can be executed after both saved runs finish. Run tests/test_recorded_results.py against that output directory to independently audit CSV/NPZ consistency and summary arithmetic.

Do not edit source, assets, config or package versions between stages. Each run manifest records their hashes and environment. Existing stage output directories are not overwritten. An interrupted run has no automatic resume; preserve it and use a new run folder. Exact bitwise reproducibility across hardware is not guaranteed; scientific agreement should be judged against stated tolerances.

9. Shareable outputs and interpretation

After comparison completes, share comparison/team_results_report.pdf, its Markdown source, comparison.csv, summary.json, failures.csv, plots and both stage recordings. Include manifest.json, used_config.json and the complete source/model package for reproduction.

The generated report uses the newly saved measurements; no previous contacts-on or contacts-off results are copied into it. It distinguishes valid-only statistics from all-finite diagnostics and states any coverage failures. Its interpretation is limited to the tested path and assembly branch.

The model is derived from CAD rather than independently certified physical ground truth. Retained home-angle mappings were previously adjusted; agreement does not independently validate that calibration. Later work should check measured hardware poses, actual joint limits, workspace/branch coverage, singularities, collision geometry and a calibrated actuator controller.

Quick commands - Windows

Create environment

```
py -3.12 -m venv .venv
```

Install dependencies

```
.venv\Scripts\python -m pip install -r requirements.txt
```

Local preflight

```
.venv\Scripts\python check_setup.py
```

Run and record

```
.venv\Scripts\python workflow.py --stage all --out results\run_1601 --preview
```

Audit saved results

```
.venv\Scripts\python tests\test_recorded_results.py results\run_1601
```